SmartGrid Sentinel

An AI-Based Load Shedding Risk Prediction and Explainable Decision Support System

Submitted By

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Abstract

Load shedding remains a major challenge in power distribution systems due to increasing electricity demand, infrastructure limitations, and environmental influences. This project presents SmartGrid Sentinel, an AI-based load shedding risk prediction and explainable decision support system developed for the Sylhet Division of Bangladesh.

The system utilizes a dataset containing 14,852 smart-grid observations collected from 41 upazilas across four districts. Three predictive models—Random Forest, XGBoost, and Bidirectional Long Short-Term Memory (Bi-LSTM)—were trained and evaluated.

Experimental results demonstrated that the Bidirectional LSTM achieved the highest predictive performance with an accuracy of 72.94%, outperforming Random Forest (72.64%) and XGBoost (72.10%).

To improve usability and transparency, an NLP-powered Explainable AI module was integrated to generate human-readable explanations and operational recommendations. A console-based deployment platform was also developed for real-time prediction and decision support.

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Chapter 1

Introduction

1.1 Background

Electricity plays a critical role in economic growth, industrial development, healthcare, and education. However, many developing countries continue to experience load shedding due to increasing electricity demand and limited generation capacity.

In Bangladesh, rapid urbanization and industrial expansion have significantly increased electricity consumption. The Sylhet Division frequently experiences grid stress caused by demand-capacity imbalance, transformer overloading, adverse weather conditions, and infrastructure limitations.

Traditional grid management approaches are reactive and often fail to provide sufficient warning before outages occur. Therefore, intelligent prediction systems are required to identify potential risks before load shedding events occur.

1.2 Problem Statement

Current power grid monitoring systems primarily focus on real-time observation rather than predictive analysis.

The absence of predictive intelligence leads to:

- Increased outage frequency
- Reduced grid reliability
- Inefficient resource allocation
- Limited decision support for operators

1.3 Objectives

The objectives of this project are:

- Predict load shedding risk using artificial intelligence models.
- Perform Upazila-level risk assessment.
- Compare Random Forest, XGBoost, and Bidirectional LSTM models.
- Generate human-readable explanations using NLP.
- Develop a deployable decision support application.

1.4 Project Contributions

The major contributions include:

- AI-driven load shedding prediction framework.
- Feature engineering using smart-grid indicators.
- Comparative evaluation of machine learning and deep learning models.
- Explainable AI recommendation system.
- Real-time deployment through a console application.

Chapter 2

Dataset Description

2.1 Dataset Overview

The SmartGrid Sentinel project utilizes a structured smart-grid dataset named

`smart_grid_dataset_v8_final.csv`.

The dataset was developed to support load shedding risk prediction at the Upazila level within the Sylhet Division of Bangladesh.

The dataset contains historical grid observations, weather conditions, electrical demand indicators, infrastructure status information, and risk labels. It serves as the primary data source for training and evaluating machine learning and deep learning models.

2.2 Dataset Characteristics

Table 2.1 presents the overall characteristics of the dataset.

Table 2.1: Dataset Summary

Property	Value
Dataset Name	smart_grid_dataset_v8_final.csv
Total Records	14,852
Total Features	17 Raw Features
Division Coverage	Sylhet
Districts	4
Upazilas	41
Time Range	April 13 – May 13, 2026
Missing Values	0
Duplicate Rows	0
Target Variable	Risk Level
Classes	Low, Medium, High

2.3 Geographical Coverage

The dataset focuses exclusively on the Sylhet Division of Bangladesh.

The four districts included are:

- Sylhet
- Habiganj
- Moulvibazar
- Sunamganj

A total of 41 Upazilas are represented within these districts, enabling localized risk prediction and regional analysis.

2.4 Feature Description

The dataset contains both categorical and numerical features.

2.4.1 Categorical Features

Table 2.2: Categorical Features

Feature	Description
division	Administrative division
district	District name
upazila	Upazila name
risk_level	Target class label

2.4.2 Weather Features

Table 2.3: Weather Features

Feature	Description
temperature	Ambient temperature (°C)
humidity	Relative humidity (%)
rainfall	Rainfall amount (mm)
wind_speed	Wind speed (km/h)

2.4.3 Grid and Infrastructure Features

Table 2.4: Electrical and Infrastructure Features

Feature	Description
electricity_demand_mw	Electricity demand
generation_capacity_mw	Available generation capacity
transformer_load_percent	Transformer loading percentage
outage_count_last_24h	Outages in previous 24 hours
grid_stability_score	Grid stability index

2.5 Target Variable

The prediction target is the `risk_level` feature.

Three classes are used:

- Low Risk
- Medium Risk
- High Risk

The objective of the prediction models is to classify each observation into one of these three categories.

2.6 Class Distribution

The dataset exhibits a moderately imbalanced distribution.

Table 2.5: Risk Level Distribution

Risk Level	Samples	Percentage
Low	4,304	29.0%
Medium	7,375	49.7%
High	3,173	21.3%

The Medium risk category represents the majority class, while the High risk category contains the fewest observations.

2.7 Data Quality Assessment

The dataset was validated before preprocessing.

The following quality checks were performed:

- Verification of missing values
- Duplicate row detection
- Range validation for numerical attributes
- District–Upazila consistency verification
- Data type validation

The final dataset contains no missing values and no duplicate records, making it suitable for predictive modeling.

2.8 Dataset Significance

The dataset combines weather conditions, infrastructure indicators, electricity demand, generation capacity, and historical outage information.

These variables provide sufficient information for learning patterns associated with load shedding risk and support the development of intelligent decision-support systems for smart-grid management.

Chapter 3

Data Preprocessing

3.1 Overview

Raw data cannot be directly used for machine learning and deep learning models. Therefore, several preprocessing operations were applied to improve data quality, generate meaningful features, and prepare the dataset for model training.

The preprocessing pipeline included:

- Data validation
- Temporal feature extraction
- Categorical encoding
- Feature engineering
- Feature scaling
- Train-test splitting
- Data persistence

3.2 Data Validation

Before preprocessing, the dataset was examined for missing values, duplicate records, and invalid entries.

The validation process confirmed:

- Missing Values: 0
- Duplicate Rows: 0
- Consistent data types across all columns

- Valid district-upazila mappings

As a result, no imputation or duplicate removal was required.

3.3 Temporal Feature Extraction

The original dataset contained a datetime column representing the observation timestamp.

Using the Pandas datetime utilities, the following temporal features were extracted:

- Hour
- Day
- Month
- Day of Week

The extracted features help capture temporal patterns associated with electricity demand and grid stress.

Table 3.1: Extracted Temporal Features

Feature	Description
hour	Hour of the day (0–23)
day	Day of the month
month	Month number
day_of_week	Day of the week (0–6)

After extraction, the original datetime column was no longer required for model training.

3.4 Categorical Encoding

Machine learning algorithms require numerical input. Therefore, categorical variables were transformed into numerical representations using Label Encoding.

The following features were encoded:

- division
- district
- upazila

- risk_level (target variable)

Separate encoders were saved to ensure consistent transformation during deployment.

Table 3.2: Saved Encoders

Encoder File	Purpose
division_encoder.pkl	Division encoding
district_encoder.pkl	District encoding
upazila_encoder.pkl	Upazila encoding
label_encoder.pkl	Risk level encoding

3.5 Feature Engineering

To improve predictive performance, additional domain-specific features were generated.

3.5.1 Demand Capacity Ratio

The demand-capacity ratio measures the relationship between electricity demand and available generation capacity.

$$DemandCapacityRatio = \frac{ElectricityDemand}{GenerationCapacity + 1} \tag{3.1}$$

A higher ratio indicates increased stress on the power grid.

3.5.2 Load Stability Ratio

The load-stability ratio combines transformer loading and grid stability information.

$$LoadStabilityRatio = \frac{TransformerLoad}{GridStabilityScore + 1} \tag{3.2}$$

Higher values indicate greater infrastructure stress.

3.5.3 Cyclic Time Encoding

Since time is cyclical, sine and cosine transformations were used.

$$HourSin = \sin\left(\frac{2\pi \times Hour}{24}\right) \tag{3.3}$$

$$HourCos = \cos\left(\frac{2\pi \times Hour}{24}\right) \tag{3.4}$$

These transformations preserve temporal continuity between 23:00 and 00:00.

3.6 Final Feature Set

After preprocessing and feature engineering, the model utilized 20 input features.

The feature set included:

- Encoded geographical features
- Weather variables
- Electrical demand indicators
- Infrastructure health indicators
- Temporal attributes
- Engineered ratio features
- Cyclic time features

3.7 Feature Scaling

Feature values exist on different numerical scales.

To normalize the feature space, MinMaxScaler was applied.

The transformation scales each feature into the range:

$$0 \leq x \leq 1$$

Benefits include:

- Faster model convergence
- Improved neural network training stability
- Reduced dominance of large-scale features

The trained scaler was saved as:

`scaler.pkl`

for deployment use.

3.8 Train-Test Split

The dataset was divided into training and testing subsets using stratified sampling.

$$Training = 80\% \tag{3.5}$$

$$Testing = 20\% \tag{3.6}$$

The split was performed using:

```
random_state = 42
stratify = y
```

This ensured reproducibility while preserving class distributions.

Table 3.3: Train-Test Split Summary

Dataset	Samples
Training Set	11,881
Testing Set	2,970
Total	14,852*

*Rounded values generated from an 80/20 split.

3.9 Data Persistence

To support efficient model training and deployment, processed data and preprocessing artifacts were saved.

Table 3.4: Saved Files

File	Purpose
X_train.npy	Training features
X_test.npy	Testing features
y_train.npy	Training labels
y_test.npy	Testing labels
scaler.pkl	Feature scaler
label_encoder.pkl	Target encoder
feature_names.pkl	Final feature names

3.10 Preprocessing Pipeline Summary

Figure 3.1 illustrates the complete preprocessing workflow.

Dataset $\rightarrow$ Validation $\rightarrow$ Temporal Extraction $\rightarrow$ Encoding $\rightarrow$ Feature Engineering $\rightarrow$ Scaling $\rightarrow$ Train-Test Split $\rightarrow$ Saved Files

The resulting processed dataset was then used to train and evaluate the Random Forest, XGBoost, and Bidirectional LSTM models described in the next chapter.

Chapter 4

Methodology

4.1 System Overview

SmartGrid Sentinel is an AI-based load shedding risk prediction and decision support system designed for the Sylhet Division of Bangladesh.

The system integrates:

- Data Preprocessing
- Feature Engineering
- Machine Learning Models
- Deep Learning Models
- Explainable AI (NLP)
- Real-Time Prediction Interface

The complete workflow transforms raw grid and weather information into actionable risk predictions and operator recommendations.

4.2 System Architecture

The overall architecture of SmartGrid Sentinel is shown in Figure 4.1.

User Input
↓
Weather + Grid Parameters
↓
Data Preprocessing
↓
Feature Engineering
↓
Feature Scaling
↓
Prediction Models
(Random Forest / XGBoost / Bidirectional LSTM)
↓
Risk Prediction
↓
NLP Explainable AI Layer
↓
AI Explanation + Recommendation
↓
SmartGrid Sentinel Report

Figure 4.1: Overall System Architecture

4.3 Machine Learning and Deep Learning Pipeline

The predictive pipeline consists of three supervised learning models trained using identical input features.

1. Random Forest
2. XGBoost
3. Bidirectional LSTM

The outputs of all models were compared to determine the best-performing approach.

4.4 Random Forest Model

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive accuracy and reduce overfitting.

The model was configured with:

- Number of Trees = 300
- Maximum Depth = 15

- Random State = 42
- Parallel Processing Enabled

Advantages include:

- Robustness to noisy data
- Strong performance on structured datasets
- Reduced overfitting compared to a single decision tree
- Feature importance analysis

4.5 XGBoost Model

Extreme Gradient Boosting (XGBoost) is a powerful ensemble technique that sequentially builds decision trees to minimize prediction errors.

Configuration:

- Number of Estimators = 500
- Maximum Depth = 8
- Learning Rate = 0.05
- Subsample = 0.8
- Feature Subsample = 0.8
- Random State = 42

Advantages include:

- Excellent predictive performance
- Regularization against overfitting
- Fast training and inference
- Ability to model complex nonlinear relationships

4.6 Bidirectional LSTM Model

The Bidirectional Long Short-Term Memory (Bi-LSTM) network was selected as the primary deep learning model.

Unlike conventional neural networks, LSTM networks are capable of learning temporal dependencies and sequential patterns.

The Bidirectional structure processes information in both forward and backward directions.

4.6.1 Network Architecture

The architecture consists of:

- Input Layer
- Bidirectional LSTM Layer (128 Units)
- Batch Normalization
- Dropout Layer (0.25)
- Bidirectional LSTM Layer (64 Units)
- Batch Normalization
- Dropout Layer (0.25)
- Dense Layer (128 Units)
- Dropout Layer (0.20)
- Output Layer (Softmax)

4.6.2 Architecture Diagram

Input (20 Features)
↓
Bi-LSTM (128)
↓
Batch Normalization
↓
Dropout (0.25)
↓
Bi-LSTM (64)
↓
Batch Normalization
↓
Dropout (0.25)
↓
Dense (128)
↓
Dropout (0.20)
↓
Softmax Output (3 Classes)

Figure 4.2: Bidirectional LSTM Architecture

4.6.3 Training Configuration

The model was trained using:

Table 4.1: Bi-LSTM Training Parameters

Parameter	Value
Optimizer	Adam
Learning Rate	0.0005
Batch Size	32
Maximum Epochs	150
Validation Split	20%
Loss Function	Sparse Categorical Crossentropy
Output Activation	Softmax

4.7 Explainable AI Module

To improve interpretability, an NLP-powered Explainable AI module was integrated into the system.

The module converts numerical predictions into human-readable explanations and recommendations.

The Explainable AI layer performs:

1. Prediction Explanation
2. Recommendation Generation
3. Natural Language Query Parsing
4. Report Generation

4.7.1 Explanation Generation

The explanation engine analyzes:

- Demand-Capacity Ratio
- Transformer Load
- Grid Stability
- Recent Outages
- Weather Conditions

and produces understandable explanations for operators.
Example:

> "Electricity demand exceeds generation capacity, indicating a potential supply deficit. Transformer loading is approaching critical limits, increasing the probability of load shedding."

4.7.2 Recommendation Generation

The recommendation engine provides operational actions based on predicted risk levels.
Examples include:

- Activate reserve generators
- Reduce industrial demand
- Monitor transformer temperatures
- Notify consumers
- Deploy maintenance teams

4.8 Natural Language Processing Component

The NLP component enables users to interact with the system using natural language.

Example queries:

Predict load shedding for Golapganj at 7 PM

Show risk level for Sylhet Sadar tonight

The parser extracts:

- Location
- Time
- User Intent

before forwarding the request to the prediction engine.

4.9 Deployment Workflow

The deployed system follows the workflow shown below:

1. User enters grid information.
2. Features are encoded and scaled.
3. Bidirectional LSTM performs prediction.
4. Risk level is determined.
5. Explainable AI generates explanations.
6. Recommendations are generated.
7. SmartGrid Sentinel report is displayed.

4.10 Methodology Summary

The proposed methodology integrates machine learning, deep learning, and natural language processing into a unified framework.

Random Forest and XGBoost provide strong baseline comparisons, while the Bidirectional LSTM captures temporal relationships within the data. The Explainable AI module enhances transparency by translating model outputs into actionable insights for grid operators.

Chapter 5

Results and Discussion

5.1 Overview

This chapter presents the experimental results obtained from the Random Forest, XG-Boost, and Bidirectional LSTM models.

All models were trained using the same preprocessed dataset and evaluated on the same testing dataset to ensure a fair comparison.

The evaluation focuses on:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix Analysis
- Training Performance

5.2 Dataset Summary

The final dataset used for training and evaluation contains:

Table 5.1: Dataset Summary

Property	Value
Total Samples	14,852
Training Samples	11,881
Testing Samples	2,970
Districts	4
Upazilas	41
Risk Classes	3

5.3 Model Performance Comparison

Table 5.2 presents the final evaluation results of all models.

Table 5.2: Performance Comparison of Models

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	72.64%	72.75%	72.64%	72.59%
XGBoost	72.10%	72.17%	72.10%	72.07%
Bidirectional LSTM	72.94%	73.63%	72.94%	72.76%

The Bidirectional LSTM achieved the highest overall performance among the evaluated models. Although the performance difference is relatively small, the deep learning model demonstrated superior capability in learning complex nonlinear relationships from the smart-grid data.

5.4 Model Comparison Visualization

Figure 5.1 illustrates the comparative performance of all models.

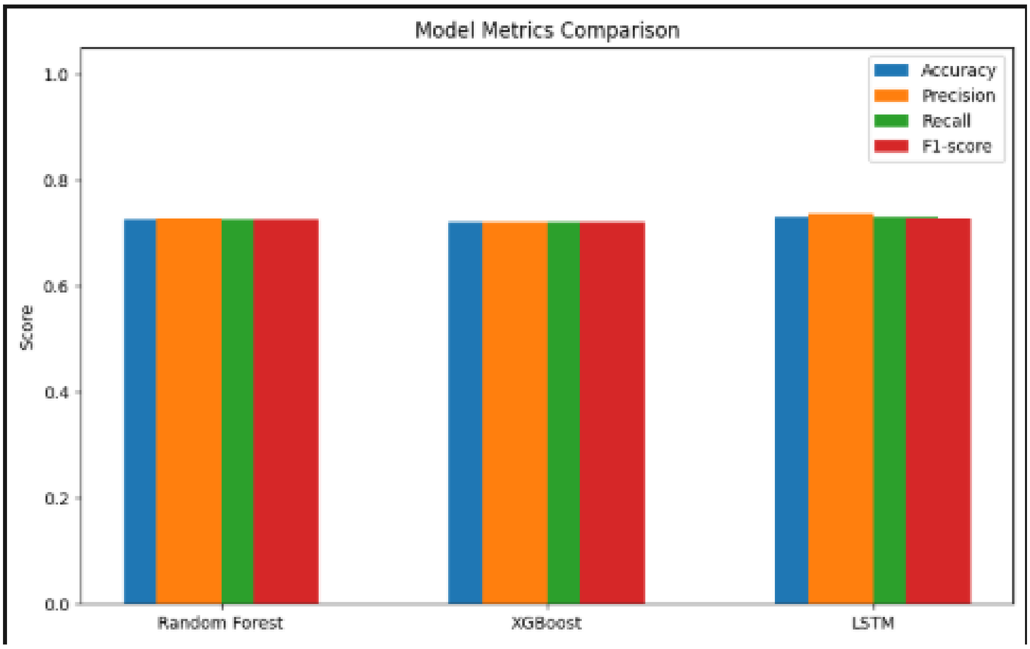

Figure 5.1: Comparison of Accuracy, Precision, Recall and F1-Score

The Bidirectional LSTM achieved the best overall balance across all evaluation metrics.

5.5 LSTM Training Analysis

The learning behavior of the Bidirectional LSTM was analyzed using training and validation metrics.

5.5.1 Training Loss

Figure 5.2 shows the loss curve during training.

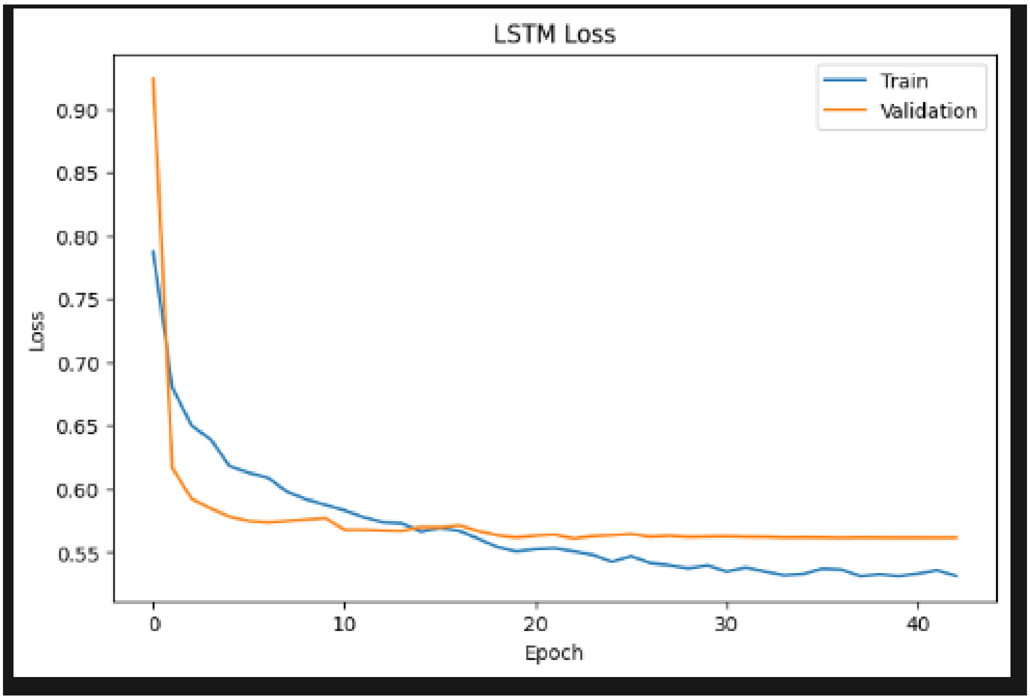

Figure 5.2: LSTM Training and Validation Loss

Both training and validation loss decreased steadily throughout the training process. The curves stabilized after approximately 20 epochs, indicating successful convergence.

5.5.2 Training Accuracy

Figure 5.3 presents the training and validation accuracy.

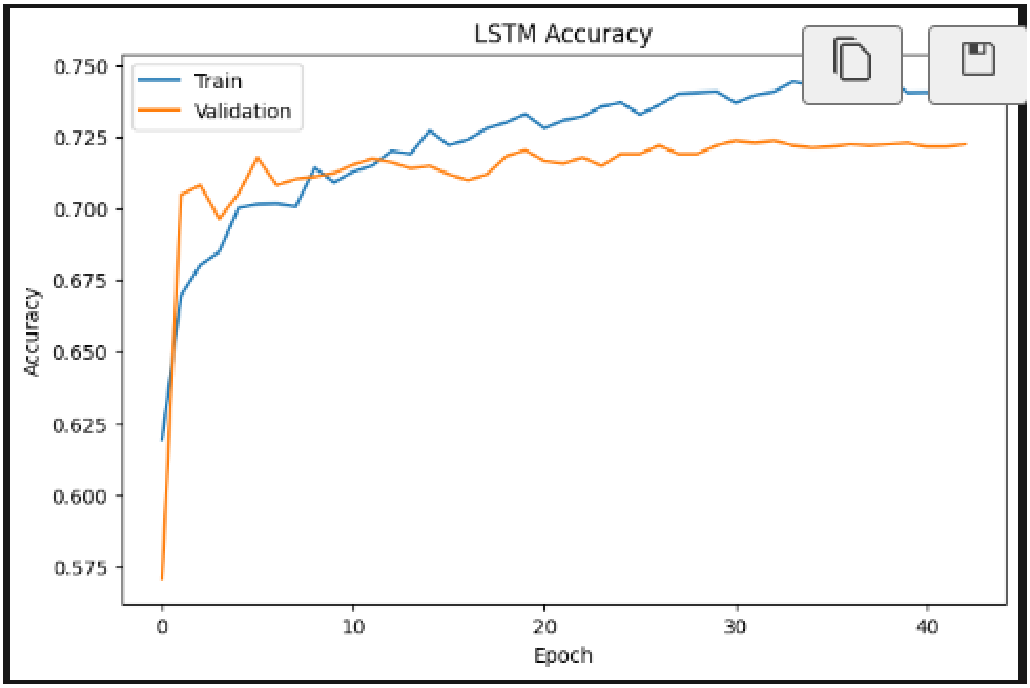

Figure 5.3: LSTM Training and Validation Accuracy

The validation accuracy remained close to the training accuracy throughout the training process. This indicates good generalization capability and limited overfitting.

5.6 Confusion Matrix Analysis

The confusion matrix of the Bidirectional LSTM model is shown in Figure 5.4.

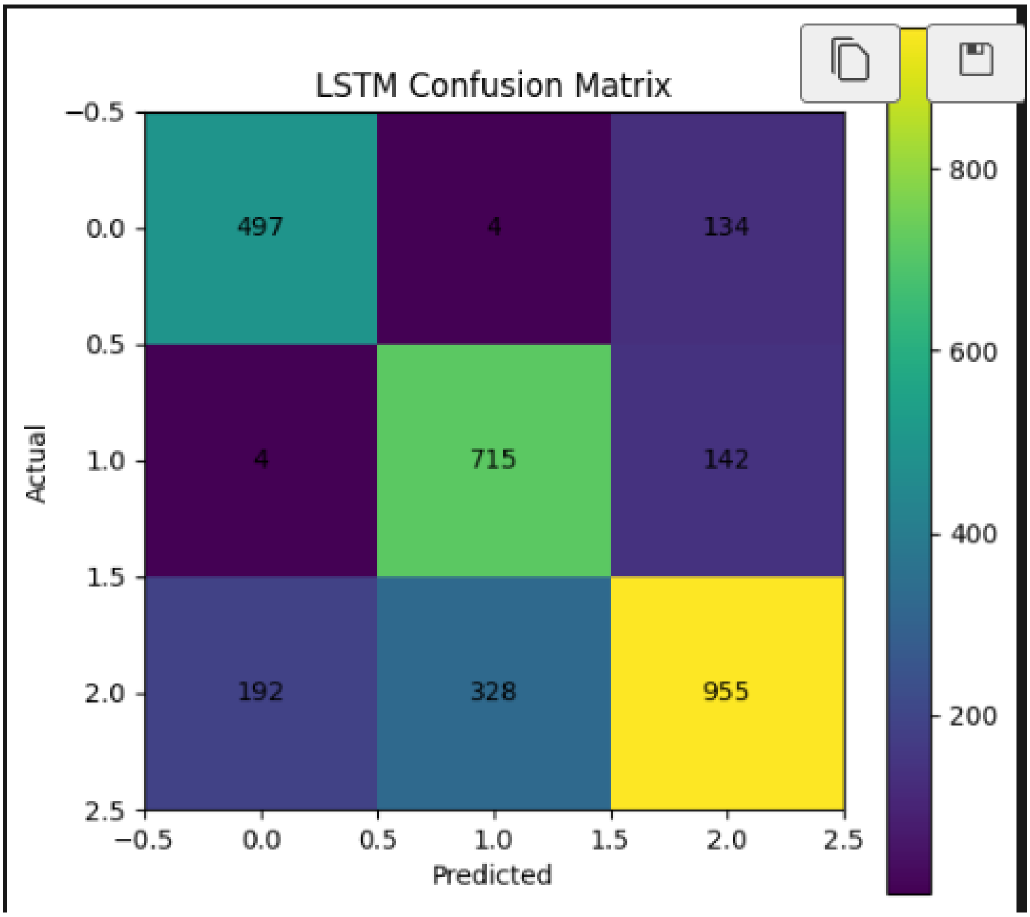

Figure 5.4: Confusion Matrix of the Bidirectional LSTM Model

Most samples are correctly classified along the diagonal of the matrix. Misclassifications primarily occur between Medium and High risk categories, which is expected because these classes often exhibit similar operating conditions.

5.7 Discussion of Results

Several important observations can be made from the experimental results:

1. Bidirectional LSTM achieved the highest predictive accuracy of 72.94%.

2. Random Forest achieved competitive performance with 72.64% accuracy.

3. XGBoost achieved 72.10% accuracy and provided stable classification performance.
4. The performance gap among the three models is relatively small, indicating that the selected features contain strong predictive information.
5. The Bidirectional LSTM benefits from its ability to learn sequential and temporal relationships that are difficult for traditional machine learning models to capture.
6. The Explainable AI module successfully transformed numerical predictions into understandable explanations and recommendations.

5.8 Practical Validation

The deployed SmartGrid Sentinel console application was tested using multiple risk scenarios.

The system successfully:

- Accepted user-provided grid information
- Generated risk predictions
- Calculated confidence scores
- Produced AI-generated explanations
- Generated operational recommendations

The results demonstrate the practical applicability of the proposed system for real-world decision support.

5.9 Summary

The experimental evaluation demonstrates that all three models provide effective load shedding risk prediction.

Among them, the Bidirectional LSTM achieved the highest overall performance with:

- Accuracy: 72.94%
- Precision: 73.63%
- Recall: 72.94%
- F1-Score: 72.76%

These results confirm the suitability of deep learning techniques for smart-grid risk prediction and support the deployment of SmartGrid Sentinel as an intelligent decision-support platform.

Chapter 6

System Implementation and Deployment

6.1 Overview

This chapter describes the implementation of SmartGrid Sentinel and its deployment as a console-based decision support system. The system integrates machine learning, deep learning, and explainable AI techniques to predict load shedding risk and provide actionable recommendations.

6.2 System Components

The SmartGrid Sentinel system consists of the following components:

- Data Preprocessing Module
- Random Forest Model
- XGBoost Model
- Bidirectional LSTM Model
- Explainable AI Module
- Console-Based Prediction Interface

6.3 Prediction Module

The prediction module is implemented in `check.py`. It loads the trained model, preprocessing objects, and user inputs to generate load shedding risk predictions.

The prediction workflow includes:

1. User input collection
2. Data preprocessing
3. Feature engineering
4. Feature scaling
5. Model prediction
6. Risk classification

6.4 User Input Parameters

The system accepts the following parameters:

- District
- Upazila
- Temperature
- Humidity
- Rainfall
- Wind Speed
- Electricity Demand
- Generation Capacity
- Transformer Load
- Grid Stability Score
- Outage Count
- Current Hour

These parameters represent the current operating condition of the electrical grid.

6.5 Explainable AI Module

The Explainable AI module is implemented in `explain.py`. Its purpose is to convert model outputs into understandable explanations and recommendations.

The module performs:

- Prediction explanation generation
- Risk-based recommendation generation
- Natural language query parsing
- SmartGrid report generation

6.6 Sample System Output

After processing the user inputs, the system generates:

- Predicted Risk Level
- Confidence Score
- AI Explanation
- Operational Recommendations

An example output is shown below:

```
Predicted Risk: High

Confidence: 92%

Reason:
- Electricity demand exceeds generation capacity.
- Transformer loading is critically high.
- Grid stability is below the safe threshold.

Recommendation:
- Activate reserve generators.
- Reduce industrial load.
- Monitor transformers closely.
```

6.7 Deployment Environment

The system was developed and tested using:

Table 6.1: Development Environment

Component	Technology
Programming Language	Python
Machine Learning	Scikit-Learn
Deep Learning	TensorFlow / Keras
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Model Storage	Pickle, H5
Operating System	Windows 10

6.8 Advantages of the System

The implemented system provides several benefits:

- Upazila-level risk prediction
- Fast prediction time
- Explainable AI outputs
- Easy-to-use interface
- Support for operational decision making
- Expandable architecture for future deployment

6.9 Future Deployment Plan

The current version operates as a console-based application. Future work will focus on developing a web-based platform using HTML, CSS, and JavaScript to provide a more user-friendly interface and wider accessibility.

6.10 Chapter Summary

This chapter presented the implementation and deployment of SmartGrid Sentinel. The developed system successfully integrates prediction models and explainable AI components into a practical decision support application for load shedding risk forecasting.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This project presented SmartGrid Sentinel, an AI-based load shedding risk prediction and decision support system developed for the Sylhet Division of Bangladesh.

The system combines machine learning, deep learning, and explainable artificial intelligence techniques to predict load shedding risk at the Upazila level. A complete pipeline was developed, including data collection, preprocessing, feature engineering, model training, evaluation, and deployment.

Three predictive models were implemented and compared:

- Random Forest
- XGBoost
- Bidirectional LSTM

Experimental results demonstrated that the Bidirectional LSTM model achieved the best overall performance with an accuracy of approximately 72.94%, outperforming the traditional machine learning models.

The project also introduced an Explainable AI module capable of generating human-readable explanations and operational recommendations. This improves transparency and supports decision-making by power grid operators.

Overall, SmartGrid Sentinel demonstrates that artificial intelligence can effectively support proactive load shedding management and improve the reliability of electrical power distribution systems.

7.2 Achievements

The major achievements of this project are summarized below:

1. Developed a complete load shedding risk prediction pipeline.
2. Built and evaluated three predictive models.
3. Achieved more than 72% prediction accuracy.
4. Implemented feature engineering techniques for improved performance.
5. Developed an NLP-based explanation and recommendation system.
6. Created a console-based decision support application.
7. Produced risk predictions at the Upazila level.

7.3 Limitations

Although the proposed system performed successfully, several limitations remain.

- The dataset is limited to the Sylhet Division.
- Electricity demand values are partially simulated.
- The current deployment is console-based.
- Real-time grid sensor integration is not yet available.
- Historical data coverage can be expanded further.

These limitations provide opportunities for future enhancement.

7.4 Future Work

Several improvements can be incorporated in future versions of SmartGrid Sentinel.

7.4.1 Nationwide Deployment

Future datasets may include all divisions, districts, and upazilas of Bangladesh to provide nationwide load shedding prediction.

7.4.2 Web-Based Platform

A web-based dashboard can be developed using:

- HTML
- CSS
- JavaScript
- Python Backend

This will improve accessibility and usability.

7.4.3 Real-Time Data Integration

Future versions may connect directly with:

- Smart meters
- SCADA systems
- Weather APIs
- Grid monitoring sensors

for real-time forecasting.

7.4.4 Advanced Deep Learning Models

Future research may investigate:

- Transformer Networks
- Temporal Fusion Transformers
- Graph Neural Networks
- Hybrid CNN-LSTM Architectures

to further improve prediction accuracy.

7.4.5 Mobile Application

A mobile application could be developed to provide instant alerts and recommendations to field operators.

7.5 Final Remarks

SmartGrid Sentinel successfully demonstrates the application of Artificial Intelligence, Deep Learning, and Explainable AI for load shedding risk prediction.

The proposed framework provides accurate forecasting, interpretable decision support, and a scalable foundation for future smart grid management systems. With further development and real-time integration, SmartGrid Sentinel has the potential to become an effective operational tool for modern power distribution networks.

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